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NORME ASD-STAN  
ASD-STAN NORM**

**ASD-STAN prEN 6115:2019**

**Edition P 10**

**June 2019**



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ICS:

Descriptors:

**ENGLISH VERSION**

**Aerospace series — Bolt — Protruding head, Short thread — Inch series**

**Luft- und Raumfahrt — Bolzen —  
Flachkopf, Kurzgewinde — Zoll-Reihe**

**Série aéronautique — Vis — tête  
cylindrique, filetage court — Série  
pouce**

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**Mechanical Domain**

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## **Foreword**

This standard was reviewed by the Domain Technical Coordinator of ASD-STAN's Mechanical Domain.

After inquiries and votes carried out in accordance with the rules of ASD-STAN defined in ASD-STAN's General Process Manual, this standard has received approval for Publication.

## **prEN 6115:2019 (E)**

### **Introduction**

This document is published at edition P10. Former editions P1 to P9 and drafts may exist for Airbus development only but without any ASD-STAN official publication. In consequence configuration management discrepancies with these unofficial documents are under Airbus responsibility.

## 1 Scope

This product standard specifies the dimensions, tolerances and the requirements of a protruding head external threaded fastener for aerospace applications.

## 2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 2424, *Aerospace series — Marking of aerospace products*

EN 4473, *Aerospace series — Aluminium pigmented coatings for fasteners — Technical specification*

EN 6116, *Aerospace series — Fasteners — External short thread, Inch series — Technical specification*<sup>1</sup>

EN 6117, *Aerospace series — Specification for lubrication of fasteners with cetyl alcohol*<sup>1</sup>

EN 6118, *Aerospace series — Process specification — Aluminium base protection for fasteners*<sup>1</sup>

ISO 8080, *Aerospace — Anodic treatment of titanium and titanium alloys — Sulphuric acid process*

AMS2700, *Passivation of Corrosion Resistant Steels*<sup>2</sup>

AMS2759/1, *Heat Treatment of Carbon and Low-Alloy Steel Parts Minimum Tensile Strength Below 220 ksi (1 517 MPa)*<sup>2</sup>

AMS4928, *Titanium Alloys Bars, Wire, Forgings, Rings, and Drawn Shapes 6Al - 4V Annealed*<sup>2</sup>

AMS4967, *Titanium Alloy, Bars, Wire, Forgings, and Rings 6.0Al - 4.0V Annealed, Heat Treatable*<sup>2</sup>

AMS5662, *Nickel Alloy, Corrosion and Heat-Resistant, Bars, Forgings, and Rings 52.5Ni - 19Cr - 3.0Mo - 5.1Cb (Nb) - 0.90Ti - 0.50Al - 18Fe Consumable Electrode or Vacuum Induction Melted 1 775 °F (968 °C) Solution Heat Treated, Precipitation-Hardenable*<sup>2</sup>

AMS5962, *Nickel Alloy, Corrosion and Heat-Resistant, Round Bars and Wire 52.5Ni - 19Cr - 3.0Mo - 5.1Cb - 0.90Ti - 0.50Al - 18Fe Consumable Electrode or Vacuum Induction Melted 1 775 °F (968 °C) Solution Treated and Work Strengthened Precipitation Hardenable*<sup>2</sup>

AMS6322, *Steel Bars, Forgings, and Rings 0.50Cr - 0.55Ni - 0.25Mo (0.38 - 0.43C) (SAE 8740)*<sup>2</sup>

AMS6325, *Steel, Bars, and Forgings 0.50Cr - 0.55Ni - 0.25Mo (0.38 - 0.43C) (SAE 8740) Heat Treated, 105 ksi (724 MPa) Tensile Strength*<sup>2</sup>

AMS6327, *Steel Bars and Forgings 0.50Cr - 0.55Ni - 0.25Mo (0.38 - 0.43C) (SAE 8740) Heat Treated, 125 ksi (862 MPa) Tensile Strength*<sup>2</sup>

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<sup>1</sup> Published as ASD-STAN Prestandard at the date of publication of this standard ([www.asd-stan.org/](http://www.asd-stan.org/)).

<sup>2</sup> Published by: Society of Automotive Engineers (SAE), 400 Commonwealth Drive, Warrendale, PA 15096-0001, USA.

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AMS6349, *Steel Bars 0.95Cr - 0.20Mo (0.38 to 0.43C) (SAE 4140) Normalized*<sup>2</sup>

AMS6382, *Steel, Bars, Forgings, and Rings 0.95Cr - 0.20Mo (0.38 - 0.43C) (SAE 4140) Annealed*<sup>2</sup>

AMS6415, *Steel, Bars, Forgings, and Tubing 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340)*<sup>2</sup>

AMS6484, *Steel, Bars, Forgings, and Tubing 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340) Normalized and Tempered*<sup>2</sup>

AMSQQP416, *Plating, Cadmium (Electrodeposited)*<sup>2</sup>

AS8879, *Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter*<sup>2</sup>

ASME B46.1, *Surface Texture (Surface Roughness, Waviness, and Lay)*<sup>3</sup>

### 3 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at <http://www.iso.org/obp>

— IEC Electropedia: available at <http://www.electropedia.org/>

### 4 Requirements

#### 4.1 Configuration, dimensions and tolerances

Configuration dimensions and tolerances shall be in accordance with Figure 1, Table 3 and Table 4.

Dimensions and tolerances for oversize fasteners shall be in accordance with Table 5, Table 6, and Table 7.

Dimensions shall apply after coating but before lubrication.

Break sharp edges 0,25 mm max radius.

#### 4.2 Mass

The calculation of the mass of an external threaded fastener shall be provided according to indications hereafter:

Add the mass of the head and threaded part (invariable mass) to the mass of the smooth part (variable mass).

Total mass of the head and threaded part:

1<sup>st</sup> mass column of Table 3.

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3 Published by: American Society of Mechanical Engineers (ASME), Two Park Avenue, New York, NY 10016-5990, USA.

Mass of the smooth part:

Multiply the value of the 2<sup>nd</sup> mass column of Table 3 (value according to the diameter code No.) by the length code No. of the external threaded fastener.

#### EXAMPLES

EN6115-4-8

Variable mass 2,98 g

Variable mass  $0,39 \text{ g} \times 8 = 3,12 \text{ g}$

Total mass 6,10 g

EN6115V4-8, EN6115T4-8, EN6115K4-8 or EN6115B4-8

Variable mass 1,53 g

Variable mass  $0,22 \text{ g} \times 8 = 1,76 \text{ g}$

Total mass 3,29 g

EN6115L4-8, EN6115P4-8 or EN6115M4-8

Variable mass 3,21 g

Variable mass  $0,42 \text{ g} \times 8 = 3,36 \text{ g}$

Total mass 6,57 g

EN6115L4X8

Variable mass 3,21 g

Variable mass  $0,45 \text{ g} \times 8 = 3,60 \text{ g}$

Total mass 6,81 g

### 4.3 Materials, finishes, lubrication and identifications

**Table 1 — Materials, finishes, lubrication and identifications**

| Code | Material                                                                                    | Finish                                                                                                                             | Lubrication                        | Recess                    |      | Dia. range        | Fastener identification                    |
|------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------|------|-------------------|--------------------------------------------|
|      |                                                                                             |                                                                                                                                    |                                    | Form                      | Code |                   |                                            |
| T    | Titanium alloy 6Al-4V according to AMS4928 or AMS4967 or equivalent<br>$R_c$ min. = 650 MPa | Sulfuric-acid anodizing according to ISO 8080                                                                                      | Cetyl alcohol according to EN 6117 | Hexagonal                 | –    | 2 to 18           | None                                       |
| V    |                                                                                             | IVD according to EN 6118                                                                                                           |                                    |                           |      |                   |                                            |
| K    |                                                                                             | Resin based Aluminium coating according to EN 4473 Type II <sup>a</sup>                                                            |                                    | 5 Lobe high torque recess | E    | 2 to 4            | A white paint identification at thread end |
| B    |                                                                                             | Sulfuric-acid anodizing according to ISO 8080 + Resin based Aluminium coating according to EN 4473 Type II <sup>a</sup> on threads |                                    |                           |      |                   | None                                       |
|      |                                                                                             |                                                                                                                                    |                                    | Hexagonal                 | –    | 2 to 4<br>5 to 18 |                                            |

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| Code                                                                                                                                                                                                                                                                                                                                                                                                                                       | Material                                                                                                                                                                                                                                               | Finish                                                                   | Lubrication                              | Recess |      | Dia.<br>range | Fastener<br>identification                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------|--------|------|---------------|---------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                        |                                                                          |                                          | Form   | Code |               |                                                   |
| -                                                                                                                                                                                                                                                                                                                                                                                                                                          | Alloy steel<br>4340 (AMS6415 or<br>AMS6484) or<br>4140 (AMS6382 or<br>AMS6349) or<br>8740 (AMS6322 or<br>AMS6325 or<br>AMS6327) or<br>equivalent<br>$R_c$ min. = 740 MPa<br>$R$ = 1 240 to<br>1 380 MPa<br>(AMS2759/1)<br>Hardness HRC 40<br>to HRC 44 | Cadmium plating<br>according to<br>AMSQQP416,<br>Type II,<br>class 2     | Cetyl alcohol<br>according to<br>EN 6117 |        | -    | 3 to 20       | A green paint<br>identification at<br>thread end  |
| L                                                                                                                                                                                                                                                                                                                                                                                                                                          | Nickel based alloy <sup>b</sup><br>(AMS5962<br>or AMS5662 + cold<br>working)<br>$R$ min. = 1 510 MPa                                                                                                                                                   | Resin based<br>Aluminium<br>according to<br>EN 4473 Type II <sup>a</sup> |                                          |        | -    | 3 to 16       | A blue paint<br>identification at<br>thread end   |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                        | IVD according to<br>EN 6118                                              |                                          |        |      |               | A black paint<br>identification at<br>thread end  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                        | Passivation<br>according to<br>AMS2700                                   |                                          |        |      |               | A yellow paint<br>identification at<br>thread end |
| <sup>a</sup> Parts produced before 1 January 2017 may be produced with EN 4473 Type I.<br><sup>b</sup> e. g. Inconel 718. Inconel® is a trade name of a product supplied by Special Metals Corporation. This information is given for the convenience of users of this standard and does not constitute an endorsement by ASD-STAN of the product named. Equivalent products may be used if they can be shown to lead to the same results. |                                                                                                                                                                                                                                                        |                                                                          |                                          |        |      |               |                                                   |



#### 4.4 Mechanical characteristics

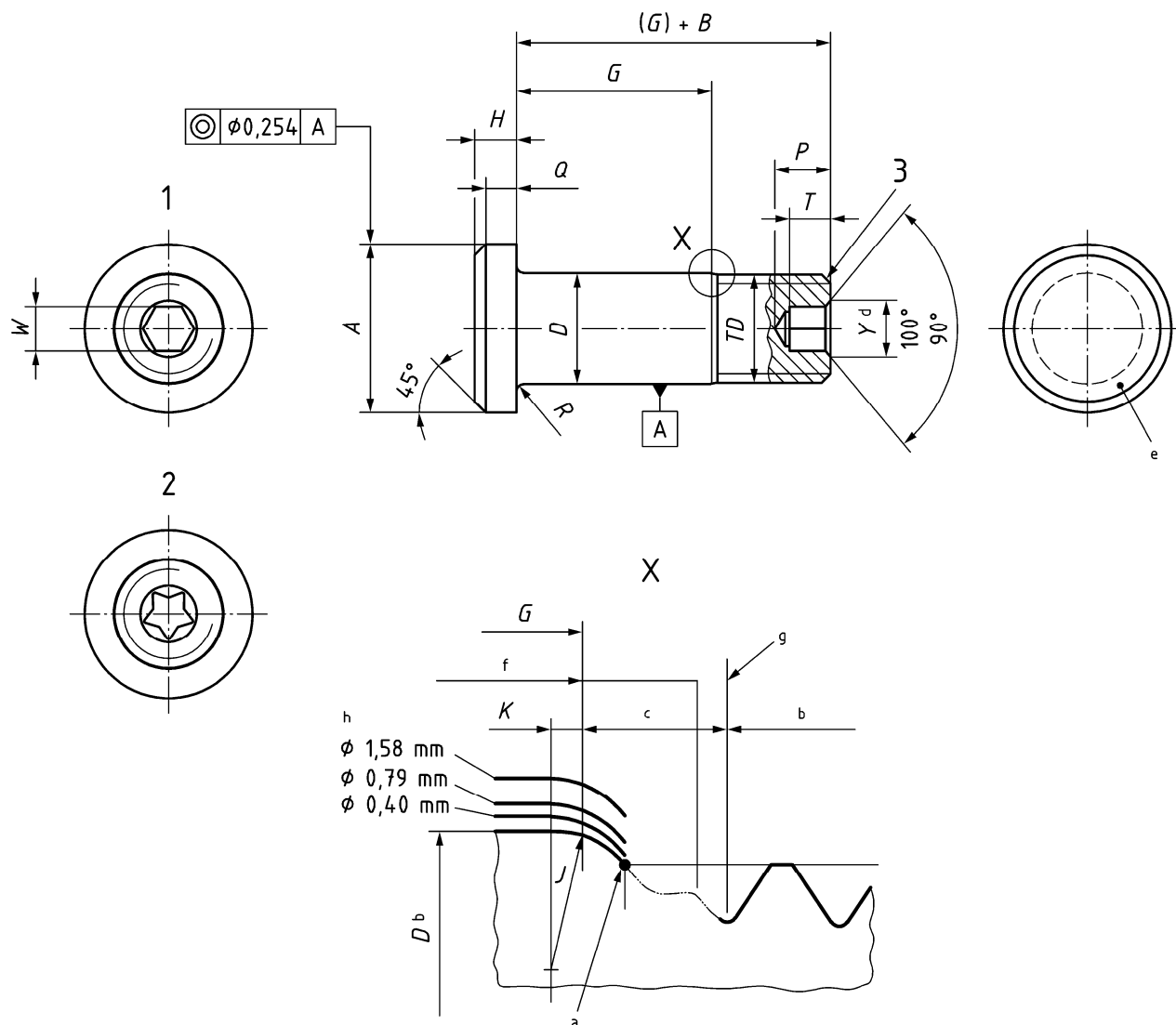
Table 2 — Mechanical characteristics

| Dia.<br>code<br>No.                                               | Min. double shear strength (N) |                |                | Min. tensile strength (N) |                |                | Max. fatigue load (N) |                |                |
|-------------------------------------------------------------------|--------------------------------|----------------|----------------|---------------------------|----------------|----------------|-----------------------|----------------|----------------|
|                                                                   | Steel alloy                    | Titanium alloy | Ni-based alloy | Steel alloy               | Titanium alloy | Ni-based alloy | Steel alloy           | Titanium alloy | Ni-based alloy |
| 2                                                                 | –                              | 17 760         | –              | –                         | 9 660          | –              | –                     | 3 170          | –              |
| 3                                                                 | 27 250                         | 23 900         | 31 500         | 17 100                    | 14 150         | 20 820         | 5 330                 | 4 670          | 6 490          |
| 3A                                                                | –                              | 32 000         | 41 600         | –                         | 17 800         | 25 060         | –                     | 6 050          | 8 770          |
| 4                                                                 | 47 150                         | 41 330         | 54 800         | 29 150                    | 25 900         | 35 490         | 9 780                 | 8 670          | 11 900         |
| 5                                                                 | 73 850                         | 64 880         | 85 100         | 46 050                    | 40 900         | 56 070         | 15 560                | 13 960         | 18 940         |
| 6                                                                 | 106 300                        | 93 320         | 123 000        | 70 050                    | 62 250         | 85 300         | 24 000                | 21 570         | 29 220         |
| 7                                                                 | 144 550                        | 127 100        | 167 300        | 94 300                    | 78 600         | 114 830        | 32 900                | 27 200         | 40 060         |
| 8                                                                 | 188 600                        | 165 760        | 218 300        | 128 100                   | 106 500        | 155 990        | 44 480                | 37 100         | 54 160         |
| 9                                                                 | 238 850                        | 209 950        | 276 230        | 161 900                   | 126 100        | 186 000        | 55 150                | 44 000         | 65 100         |
| 10                                                                | 294 900                        | 259 330        | 341 300        | 205 050                   | 170 600        | 220 000        | 69 830                | 59 900         | 74 800         |
| 12                                                                | 424 350                        | 373 200        | 491 130        | 297 600                   | 247 320        | 362 390        | 101 420               | 86 290         | 123 500        |
| 14                                                                | 573 800                        | 493 990        | 664 100        | 413 700                   | 302 000        | 498 740        | 140 560               | 112 020        | 169 967        |
| 16                                                                | 749 500                        | 625 970        | 867 450        | 536 000                   | 394 000        | 587 500        | 182 380               | 141 950        | 217 400        |
| 18                                                                | 951 900                        | 769 170        | –              | 684 150                   | 509 730        | –              | 218 640               | 174 420        | –              |
| 20                                                                | 1 178 800                      | –              | –              | 898 500                   | –              | –              | 305 600               | –              | –              |
| 22                                                                | –                              | –              | –              | –                         | –              | –              | –                     | –              | –              |
| NOTE Minimum fatigue loads are equal to 10 % of maximum loads.    |                                |                |                |                           |                |                |                       |                |                |
| NOTE Mechanical characteristics also apply to oversize fasteners. |                                |                |                |                           |                |                |                       |                |                |

#### 4.5 General characteristics

The surface condition shall be according to ASME B46.1.

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### Key

- 1 Hexagonal recess code “-” (see Table 1)
- 2 5 Lobe high torque recess code “E” (see Table 1)
- 3 Chamfer S

NOTE The maximum thread runout and functionality for 1<sup>st</sup> oversize is increased by 0,25 mm, for 2<sup>nd</sup> oversize by 0,5 mm, and for 3<sup>rd</sup> oversize by 1 mm.

- a For nominal diameter: The diameter measured at point I shall be less than or equal to the maximum diameter *TD* for nominal diameter. For oversizes diameters: Maximum diameter at point I shall be incremented by 0,40 mm, 0,79 mm and 1,58 mm for respectively 1<sup>st</sup>, 2<sup>nd</sup>, and 3<sup>rd</sup> oversize. The *TD* diameter stays the same as for nominal diameter.
- b Only for B and C coded fasteners, threads shall be coated with aluminium coating according to EN 4473
- c Only for B and C coded fasteners, overspray of aluminium coating according to EN 4473 is permissible in this area.
- d Valid for hexagonal recess.
- e Marking
- f Thread functionality 1,0 pitch
- g Thread runout max. 1,5 pitch
- h Oversizes

**Figure 1 — Configuration, dimensions and tolerances**

Table 3 — Dimensions, tolerances and mass

Dimensions in millimetres

| Dia.<br>code<br>No.                                                 | Nominal<br>shank<br>diameter | Thread <sup>a</sup><br>UNJF-3A modified<br>(inch) | <i>A</i>       | <i>B</i><br>Ref. | <i>D</i>          |                  | <i>TD</i>        | <i>H</i>       | <i>R</i>     |
|---------------------------------------------------------------------|------------------------------|---------------------------------------------------|----------------|------------------|-------------------|------------------|------------------|----------------|--------------|
|                                                                     |                              |                                                   |                |                  | T, B or P<br>code | Other<br>code    |                  |                |              |
| 2                                                                   | 3,97                         | 0.1640-32 <sup>b</sup>                            | 8,17<br>7,77   | 7,11             | 4,153<br>4,140    | 4,153<br>4,128   | 4,051<br>3,988   | 1,65<br>1,40   | 0,64<br>0,38 |
| 3                                                                   | 4,76                         | 0.1900-32                                         | 9,57<br>9,07   | 7,37             | 4,813<br>4,800    | 4,813<br>4,788   | 4,673<br>4,597   | 1,88<br>1,63   |              |
| 3A                                                                  | 5,56                         | 0.2160-28                                         | 10,41<br>9,91  | 7,75             | 5,542<br>5,529    | 5,542<br>5,517   | 5,334<br>5,258   | 2,06<br>1,80   |              |
| 4                                                                   | 6,35                         | 0.2500-28                                         | 11,17<br>10,54 | 8,13             | 6,337<br>6,324    | 6,337<br>6,312   | 6,197<br>6,121   | 2,28<br>2,03   |              |
| 5                                                                   | 7,94                         | 0.3125-24                                         | 12,82<br>12,07 | 9,65             | 7,925<br>7,911    | 7,925<br>7,899   | 7,772<br>7,670   | 2,84<br>2,59   | 0,76<br>0,51 |
| 6                                                                   | 9,52                         | 0.3750-24                                         | 15,24<br>14,35 | 10,67            | 9,512<br>9,500    | 9,512<br>9,487   | 9,347<br>9,245   | 3,55<br>3,30   |              |
| 7                                                                   | 11,11                        | 0.4375-20                                         | 17,17<br>16,28 | 12,32            | 11,099<br>11,087  | 11,099<br>11,074 | 10,947<br>10,820 | 4,06<br>3,81   |              |
| 8                                                                   | 12,70                        | 0.5000-20                                         | 19,56<br>18,67 | 13,33            | 12,687<br>12,674  | 12,687<br>12,662 | 12,522<br>12,395 | 4,77<br>4,52   |              |
| 9                                                                   | 14,29                        | 0.5625-18                                         | 22,27<br>21,39 | 15,24            | 14,262<br>14,249  | 14,262<br>14,237 | 14,097<br>13,970 | 5,33<br>5,08   | 1,01<br>0,64 |
| 10                                                                  | 15,88                        | 0.6250-18                                         | 24,20<br>23,32 | 16,26            | 15,849<br>15,836  | 15,849<br>15,824 | 15,697<br>15,545 | 6,04<br>5,79   |              |
| 12                                                                  | 19,05                        | 0.7500-16                                         | 29,21<br>28,19 | 22,73            | 19,024<br>19,011  | 19,024<br>18,999 | 18,872<br>18,720 | 8,51<br>8,13   | 1,14<br>0,76 |
| 14                                                                  | 22,22                        | 0.8750-14                                         | 33,78<br>32,76 | 25,40            | 22,200<br>22,187  | 22,200<br>22,174 | 22,047<br>21,869 | 9,78<br>9,40   | 1,27<br>0,89 |
| 16                                                                  | 25,40                        | 1.0000-12                                         | 38,35<br>37,34 | 29,46            | 25,375<br>25,362  | 25,375<br>25,349 | 25,222<br>25,044 | 11,05<br>10,67 | 1,52<br>1,14 |
| 18                                                                  | 28,58                        | 1.1250-12                                         | 43,05<br>42,04 | 33,15            | 28,550<br>28,537  | 28,550<br>28,524 | 28,372<br>28,194 | 12,44<br>12,06 | 1,78<br>1,40 |
| 20                                                                  | 31,75                        | 1.2500-12                                         | 48,50<br>47,50 | 36,54            | 31,725<br>31,712  | 31,725<br>31,700 | 31,532<br>31,232 | 14,46<br>14,06 | 1,95<br>1,85 |
| <sup>a</sup> Thread according to AS8879 except diameter <i>TD</i> . |                              |                                                   |                |                  |                   |                  |                  |                |              |
| <sup>b</sup> Thread UNJC-3A.                                        |                              |                                                   |                |                  |                   |                  |                  |                |              |

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Table 3 — Dimensions, tolerances and mass (continued)

Dimensions in millimetres

| Dia.<br>code<br>No.                                                                                                     | Nominal<br>shank<br>dia. | Thread <sup>a</sup><br>UNJF-3A<br>modified<br>(inch) | Q              | S <sup>c</sup><br>Ref. | Hexagonal recess |                |                |                | 5 Lobe high<br>torque recess |           |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------|----------------|------------------------|------------------|----------------|----------------|----------------|------------------------------|-----------|
|                                                                                                                         |                          |                                                      |                |                        | P<br>max.        | W              | T              | Y              | T<br>min.                    | P<br>max. |
| 2                                                                                                                       | 3,97                     | 0.1640-32 <sup>b</sup>                               | 1,01<br>0,51   | 0,79                   | 3,43             | 2,05<br>2,01   | 2,54<br>2,04   | 2,64<br>2,39   | 1,73                         | 3,03      |
| 3                                                                                                                       | 4,76                     | 0.1900-32                                            | 1,27<br>0,64   |                        |                  |                |                | 3,02<br>2,64   | 1,77                         | 3,10      |
| 3A                                                                                                                      | 5,56                     | 0.2160-28                                            | 1,65<br>0,77   |                        |                  |                |                | 1,80           | 3,30                         |           |
| 4                                                                                                                       | 6,35                     | 0.2500-28                                            | 1,65<br>0,89   |                        | 3,78             | 2,46<br>2,41   | 2,79<br>2,29   | 3,61<br>3,10   | 1,97                         | 3,50      |
| 5                                                                                                                       | 7,94                     | 0.3125-24                                            | 2,03<br>1,01   | 1,19                   | 4,14             | 3,29<br>3,23   | 2,99<br>2,49   | 4,57<br>4,06   | -                            | -         |
| 6                                                                                                                       | 9,52                     | 0.3750-24                                            | 2,79<br>1,39   |                        | 5,26             | 4,11<br>4,02   | 3,68<br>3,18   | 5,51<br>5,00   |                              |           |
| 7                                                                                                                       | 11,11                    | 0.4375-20                                            | 3,30<br>1,90   |                        | 5,97             | 4,90<br>4,81   | 4,36<br>3,86   | 6,43<br>5,92   |                              |           |
| 8                                                                                                                       | 12,70                    | 0.5000-20                                            |                |                        | 6,88             | 5,69<br>5,61   | 5,05<br>4,55   | 7,34<br>6,83   |                              |           |
| 9                                                                                                                       | 14,29                    | 0.5625-18                                            | 4,06<br>2,66   | 1,59                   | 8,08             | 6,49<br>6,40   | 5,97<br>5,47   | 8,28<br>7,77   |                              |           |
| 10                                                                                                                      | 15,88                    | 0.6250-18                                            | 4,44<br>3,04   |                        |                  |                |                |                |                              |           |
| 12                                                                                                                      | 19,05                    | 0.7500-16                                            | 5,97<br>4,57   |                        | 9,86             | 8,09<br>8,00   | 7,34<br>6,84   | 10,11<br>9,60  |                              |           |
| 14                                                                                                                      | 22,22                    | 0.8750-14                                            | 7,24<br>5,84   | 1,98                   | 12,09            | 9,70<br>9,60   | 8,94<br>8,44   | 11,96<br>11,46 |                              |           |
| 16                                                                                                                      | 25,40                    | 1.0000-12                                            | 8,51<br>7,11   |                        | 15,80            | 12,95<br>12,83 | 11,68<br>11,18 | 15,70<br>15,19 |                              |           |
| 18                                                                                                                      | 28,58                    | 1.1250-12                                            | 9,78<br>8,38   |                        | 18,14            | 14,54<br>14,41 | 13,05<br>12,55 | 17,60<br>17,09 |                              |           |
| 20                                                                                                                      | 31,75                    | 1.2500-12                                            | 11,56<br>10,16 | 2,10                   | 19,75            | 16,70<br>16,13 | 14,43<br>13,93 | 19,27<br>19,07 |                              |           |
| <sup>a</sup> Thread according to AS8879 except diameter TD.<br><sup>b</sup> Thread UNJC-3A.<br><sup>c</sup> 37° to 45°. |                          |                                                      |                |                        |                  |                |                |                |                              |           |

Table 3 — Dimensions, tolerances and mass (continued)

Dimensions in millimetres

| Dia.<br>code<br>No.                                                                                 | Nominal<br>shank dia. | Thread <sup>a</sup><br>UNJF-3A<br>modified<br>(inch) | Detail X       |                  | Mass Ref. (g)   |                   |                       |                |                   |                   |
|-----------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------|----------------|------------------|-----------------|-------------------|-----------------------|----------------|-------------------|-------------------|
|                                                                                                     |                       |                                                      | <i>J</i>       | <i>K</i><br>max. | Head and thread |                   |                       | Smooth part    |                   |                   |
|                                                                                                     |                       |                                                      |                |                  | Steel<br>alloy  | Titanium<br>alloy | Ni-<br>based<br>alloy | Steel<br>alloy | Titanium<br>alloy | Ni-based<br>alloy |
| 2                                                                                                   | 3,97                  | 0.1640-32 <sup>b</sup>                               | 2,16<br>1,65   | 0,33             | –               | 0,62              | –                     | –              | 0,10              | –                 |
| 3                                                                                                   | 4,76                  | 0.1900-32                                            | 2,67<br>1,78   | 0,41             | 1,60            | 0,87              | 1,72                  | 0,22           | 0,13              | 0,24              |
| 3A                                                                                                  | 5,56                  | 0.2160-28                                            | 2,92<br>2,41   | 0,48             | –               | 1,25              | 2,32                  | –              | 0,17              | 0,31              |
| 4                                                                                                   | 6,35                  | 0.2500-28                                            | 3,68<br>3,18   | 0,53             | 2,98            | 1,53              | 3,21                  | 0,39           | 0,22              | 0,42              |
| 5                                                                                                   | 7,94                  | 0.3125-24                                            | 4,45<br>3,94   | 0,66             | 5,10            | 2,81              | 5,49                  | 0,61           | 0,35              | 0,66              |
| 6                                                                                                   | 9,52                  | 0.3750-24                                            | 5,97<br>5,46   | 0,76             | 8,77            | 4,80              | 9,44                  | 0,88           | 0,51              | 0,95              |
| 7                                                                                                   | 11,11                 | 0.4375-20                                            | 7,75<br>7,24   | 0,89             | 13,63           | 7,19              | 14,67                 | 1,20           | 0,69              | 1,29              |
| 8                                                                                                   | 12,70                 | 0.5000-20                                            | 9,14<br>8,63   | 0,99             | 19,81           | 10,33             | 21,31                 | 1,56           | 0,90              | 1,68              |
| 9                                                                                                   | 14,29                 | 0.5625-18                                            | 9,78<br>9,27   |                  | 29,22           | 16,12             | 31,44                 | 1,97           | 1,14              | 2,12              |
| 10                                                                                                  | 15,88                 | 0.6250-18                                            | 10,03<br>9,40  | 1,04             | 40,31           | 21,82             | 43,37                 | 2,44           | 1,41              | 2,63              |
| 12                                                                                                  | 19,05                 | 0.7500-16                                            | 10,29<br>9,53  | 1,12             | 97,08           | 55,30             | 104,45                | 3,52           | 2,01              | 3,79              |
| 14                                                                                                  | 22,22                 | 0.8750-14                                            | 10,41<br>9,65  | 1,14             | 152,00          | 86,58             | 163,54                | 4,79           | 2,73              | 5,15              |
| 16                                                                                                  | 25,40                 | 1.0000-12                                            | 11,18<br>10,41 |                  | 225,00          | 128,16            | 242,09                | 6,26           | 3,57              | 6,74              |
| 18                                                                                                  | 28,58                 | 1.1250-12                                            | 12,18<br>11,18 | 1,19             | 326,00          | 149,94            | 278,21                | 7,91           | 4,50              | 8,35              |
| 20                                                                                                  | 31,75                 | 1.2500-12                                            | 12,95<br>12,18 |                  | 379,14          | 214,51            | 398,02                | 9,83           | 5,56              | 10,32             |
| <sup>a</sup> Thread according to AS8879 except diameter <i>TD</i> .<br><sup>b</sup> Thread UNJC-3A. |                       |                                                      |                |                  |                 |                   |                       |                |                   |                   |

Table 4 — Dimensions and tolerances

Dimensions in millimetres

| Length code<br>No. <sup>a</sup> | <i>G</i><br>±0,13 | Length ( <i>G</i> + <i>B</i> ref.) ± 0,25 |       |       |       |       |       |       |       |
|---------------------------------|-------------------|-------------------------------------------|-------|-------|-------|-------|-------|-------|-------|
|                                 |                   | 2                                         | 3     | 3A    | 4     | 5     | 6     | 7     | 8     |
| 1                               | 1,59              | 8,70                                      | 8,95  | 9,34  | 9,72  | 11,24 | –     | –     | –     |
| 2                               | 3,18              | 10,29                                     | 10,54 | 10,93 | 11,31 | 12,83 | 13,85 | –     | –     |
| 3                               | 4,76              | 11,87                                     | 12,12 | 12,51 | 12,89 | 14,41 | 15,43 | 17,08 | –     |
| 4                               | 6,35              | 13,46                                     | 13,71 | 14,10 | 14,48 | 16,00 | 17,02 | 18,67 | 19,68 |
| 5                               | 7,94              | 15,05                                     | 15,30 | 15,69 | 16,07 | 17,59 | 18,61 | 20,26 | 21,27 |
| 6                               | 9,52              | 16,63                                     | 16,88 | 17,27 | 17,65 | 19,17 | 20,19 | 21,84 | 22,85 |
| 7                               | 11,11             | 18,22                                     | 18,47 | 18,86 | 19,24 | 20,76 | 21,78 | 23,43 | 24,44 |
| 8                               | 12,70             | 19,81                                     | 20,06 | 20,45 | 20,83 | 22,35 | 23,37 | 25,02 | 26,03 |
| 9                               | 14,29             | 21,40                                     | 21,65 | 22,04 | 22,42 | 23,94 | 24,96 | 26,61 | 27,62 |
| 10                              | 15,88             | 22,99                                     | 23,24 | 23,63 | 24,01 | 25,53 | 26,55 | 28,20 | 29,21 |
| 11                              | 17,46             | 24,57                                     | 24,82 | 25,21 | 25,59 | 27,11 | 28,13 | 29,78 | 30,79 |
| 12                              | 19,05             | 26,16                                     | 26,41 | 26,80 | 27,18 | 28,70 | 29,72 | 31,37 | 32,38 |
| 13                              | 20,64             | 27,75                                     | 28,00 | 28,39 | 28,77 | 30,29 | 31,31 | 32,96 | 33,97 |
| 14                              | 22,22             | 29,33                                     | 29,58 | 29,97 | 30,35 | 31,87 | 32,89 | 34,54 | 35,55 |
| 15                              | 23,81             | 30,92                                     | 31,17 | 31,56 | 31,94 | 33,46 | 34,48 | 36,13 | 37,14 |
| 16                              | 25,40             | 32,51                                     | 32,76 | 33,15 | 33,53 | 35,05 | 36,07 | 37,72 | 38,73 |
| 17                              | 26,99             | 34,10                                     | 34,35 | 34,74 | 35,12 | 36,64 | 37,66 | 39,31 | 40,32 |
| 18                              | 28,58             | 35,69                                     | 35,94 | 36,33 | 36,71 | 38,23 | 39,25 | 40,90 | 41,91 |
| 19                              | 30,16             | 37,27                                     | 37,52 | 37,91 | 38,29 | 39,81 | 40,83 | 42,48 | 43,49 |
| 20                              | 31,75             | 38,86                                     | 39,11 | 39,50 | 39,88 | 41,40 | 42,42 | 44,07 | 45,08 |
| 21                              | 33,34             | 40,45                                     | 40,70 | 41,09 | 41,47 | 42,99 | 44,01 | 45,66 | 46,67 |
| 22                              | 34,92             | 42,03                                     | 42,28 | 42,67 | 43,05 | 44,57 | 45,59 | 47,24 | 48,25 |
| 23                              | 36,51             | 43,62                                     | 43,87 | 44,26 | 44,64 | 46,16 | 47,18 | 48,83 | 49,84 |
| 24                              | 38,10             | 45,21                                     | 45,46 | 45,85 | 46,23 | 47,75 | 48,77 | 50,42 | 51,43 |
| 25                              | 39,69             | 46,80                                     | 47,05 | 47,44 | 47,82 | 49,34 | 50,36 | 52,01 | 53,02 |
| 26                              | 41,28             | 48,39                                     | 48,64 | 49,03 | 49,41 | 50,93 | 51,95 | 53,60 | 54,61 |
| 27                              | 42,86             | 49,97                                     | 50,22 | 50,61 | 50,99 | 52,51 | 53,53 | 55,18 | 56,19 |
| 28                              | 44,45             | 51,56                                     | 51,81 | 52,20 | 52,58 | 54,10 | 55,12 | 56,77 | 57,78 |
| 29                              | 46,04             | 53,15                                     | 53,40 | 53,79 | 54,17 | 55,69 | 56,71 | 58,36 | 59,37 |
| 30                              | 47,62             | 54,73                                     | 54,98 | 55,37 | 55,75 | 57,27 | 58,29 | 59,94 | 60,95 |
| 31                              | 49,21             | 56,32                                     | 56,57 | 56,96 | 57,34 | 58,86 | 59,88 | 61,53 | 62,54 |
| 32                              | 50,80             | 57,91                                     | 58,16 | 58,55 | 58,93 | 60,45 | 61,47 | 63,12 | 64,13 |

| Length code<br>No. <sup>a</sup>                                                            | <i>G</i><br>±0,13 | Length ( <i>G</i> + <i>B</i> ref.) ± 0,25 |        |        |        |        |        |        |        |
|--------------------------------------------------------------------------------------------|-------------------|-------------------------------------------|--------|--------|--------|--------|--------|--------|--------|
|                                                                                            |                   | 2                                         | 3      | 3A     | 4      | 5      | 6      | 7      | 8      |
| 34                                                                                         | 53,98             | 61,09                                     | 61,34  | 61,73  | 62,11  | 63,63  | 64,65  | 66,30  | 67,31  |
| 36                                                                                         | 57,15             | 64,26                                     | 64,51  | 64,90  | 65,28  | 66,80  | 67,82  | 69,47  | 70,48  |
| 38                                                                                         | 60,32             | 67,43                                     | 67,68  | 68,07  | 68,45  | 69,97  | 70,99  | 72,64  | 73,65  |
| 40                                                                                         | 63,50             | 70,61                                     | 70,86  | 71,25  | 71,63  | 73,15  | 74,17  | 75,82  | 76,83  |
| 42                                                                                         | 66,68             | 73,79                                     | 74,04  | 74,43  | 74,81  | 76,33  | 77,35  | 79,00  | 80,01  |
| 44                                                                                         | 69,85             | 76,96                                     | 77,21  | 77,60  | 77,98  | 79,50  | 80,52  | 82,17  | 83,18  |
| 46                                                                                         | 73,02             | 80,13                                     | 80,38  | 80,77  | 81,15  | 82,67  | 83,69  | 85,34  | 86,35  |
| 48                                                                                         | 76,20             | 83,31                                     | 83,56  | 83,95  | 84,33  | 85,85  | 86,87  | 88,52  | 89,53  |
| 50                                                                                         | 79,38             | 86,49                                     | 86,74  | 87,13  | 87,51  | 89,03  | 90,05  | 91,70  | 92,71  |
| 52                                                                                         | 82,55             | 89,66                                     | 89,91  | 90,30  | 90,68  | 92,20  | 93,22  | 94,87  | 95,88  |
| 54                                                                                         | 85,72             | 92,83                                     | 93,08  | 93,47  | 93,85  | 95,37  | 96,39  | 98,04  | 99,05  |
| 56                                                                                         | 88,90             | 96,01                                     | 96,26  | 96,65  | 97,03  | 98,55  | 99,57  | 101,22 | 102,23 |
| 58                                                                                         | 92,08             | 99,19                                     | 99,44  | 99,83  | 100,21 | 101,73 | 102,75 | 104,40 | 105,41 |
| 60                                                                                         | 95,25             | 102,36                                    | 102,61 | 103,00 | 103,38 | 104,90 | 105,92 | 107,57 | 108,58 |
| 62                                                                                         | 98,43             | 105,54                                    | 105,80 | 106,18 | 106,56 | 108,08 | 109,10 | 110,75 | 111,76 |
| 64                                                                                         | 101,60            | 108,71                                    | 108,97 | 109,35 | 109,73 | 111,25 | 112,27 | 113,92 | 114,93 |
| 66                                                                                         | 104,78            | 111,89                                    | 112,15 | 112,53 | 112,91 | 114,43 | 115,45 | 117,10 | 118,11 |
| 68                                                                                         | 107,95            | 115,06                                    | 115,32 | 115,70 | 116,08 | 117,60 | 118,62 | 120,27 | 121,28 |
| 70                                                                                         | 111,13            | 118,24                                    | 118,50 | 118,88 | 119,26 | 120,78 | 121,80 | 123,45 | 124,46 |
| 72                                                                                         | 114,30            | 121,41                                    | 121,67 | 122,05 | 122,43 | 123,95 | 124,97 | 126,62 | 127,63 |
| 74                                                                                         | 117,48            | 124,59                                    | 124,85 | 125,23 | 125,61 | 127,13 | 128,15 | 129,80 | 130,81 |
| 76                                                                                         | 120,65            | 127,76                                    | 128,02 | 128,40 | 128,78 | 130,30 | 131,32 | 132,97 | 133,98 |
| 78                                                                                         | 123,83            | 130,94                                    | 131,20 | 131,58 | 131,96 | 133,48 | 134,50 | 136,15 | 137,16 |
| 80                                                                                         | 127,00            | 134,11                                    | 134,37 | 134,75 | 135,13 | 136,65 | 137,67 | 139,32 | 140,33 |
| <sup>a</sup> Intermediate grip lengths may be purchased in 1,59 mm increment if necessary. |                   |                                           |        |        |        |        |        |        |        |

Table 4 — Dimensions and tolerances (continued)

Dimensions in millimetres

| Length code<br>No. <sup>a</sup> | G<br>±0,13 | Length (G + B ref.) ± 0,25 |       |       |       |       |       |       |
|---------------------------------|------------|----------------------------|-------|-------|-------|-------|-------|-------|
|                                 |            | 9                          | 10    | 12    | 14    | 16    | 18    | 20    |
| 1                               | 1,59       | –                          | –     | –     | –     | –     | –     | –     |
| 2                               | 3,18       | –                          | –     | –     | –     | –     | –     | –     |
| 3                               | 4,76       | –                          | –     | –     | –     | –     | –     | –     |
| 4                               | 6,35       | –                          | –     | –     | –     | –     | –     | –     |
| 5                               | 7,94       | 23,18                      | –     | –     | –     | –     | –     | –     |
| 6                               | 9,52       | 24,76                      | 25,77 | –     | –     | –     | –     | –     |
| 7                               | 11,11      | 26,35                      | 27,36 | 33,84 | –     | –     | –     | –     |
| 8                               | 12,70      | 27,94                      | 28,95 | 35,43 | 38,10 | –     | –     | –     |
| 9                               | 14,29      | 29,53                      | 30,54 | 37,02 | 39,69 | 43,75 | –     | –     |
| 10                              | 15,88      | 31,12                      | 32,13 | 38,61 | 41,28 | 45,34 | 49,03 | –     |
| 11                              | 17,46      | 32,70                      | 33,71 | 40,19 | 42,86 | 46,92 | 50,61 | 54,00 |
| 12                              | 19,05      | 34,29                      | 35,30 | 41,78 | 44,45 | 48,51 | 52,20 | 55,59 |
| 13                              | 20,64      | 35,88                      | 36,89 | 43,37 | 46,04 | 50,10 | 53,79 | 57,18 |
| 14                              | 22,22      | 37,46                      | 38,47 | 44,95 | 47,62 | 51,68 | 55,37 | 58,76 |
| 15                              | 23,81      | 39,05                      | 40,06 | 46,54 | 49,21 | 53,27 | 56,96 | 60,35 |
| 16                              | 25,40      | 40,64                      | 41,65 | 48,13 | 50,80 | 54,86 | 58,55 | 61,94 |
| 17                              | 26,99      | 42,23                      | 43,24 | 49,72 | 52,39 | 56,45 | 60,14 | 63,53 |
| 18                              | 28,58      | 43,82                      | 44,83 | 51,31 | 53,98 | 58,04 | 61,73 | 65,12 |
| 19                              | 30,16      | 45,40                      | 46,41 | 52,89 | 55,56 | 59,62 | 63,31 | 66,70 |
| 20                              | 31,75      | 46,99                      | 48,00 | 54,48 | 57,15 | 61,21 | 64,90 | 68,29 |
| 21                              | 33,34      | 48,58                      | 49,59 | 56,07 | 58,74 | 62,80 | 66,49 | 69,88 |
| 22                              | 34,92      | 50,16                      | 51,17 | 57,65 | 60,32 | 64,38 | 68,07 | 71,46 |
| 23                              | 36,51      | 51,75                      | 52,76 | 59,24 | 61,91 | 65,97 | 69,66 | 73,05 |
| 24                              | 38,10      | 53,34                      | 54,35 | 60,83 | 63,50 | 67,56 | 71,25 | 74,64 |
| 25                              | 39,69      | 54,93                      | 55,94 | 62,42 | 65,09 | 69,15 | 72,84 | 76,23 |
| 26                              | 41,28      | 56,52                      | 57,53 | 64,01 | 66,68 | 70,74 | 74,43 | 77,82 |
| 27                              | 42,86      | 58,10                      | 59,11 | 65,59 | 68,26 | 72,32 | 76,01 | 79,40 |
| 28                              | 44,45      | 59,69                      | 60,70 | 67,18 | 69,85 | 73,91 | 77,60 | 80,99 |
| 29                              | 46,04      | 61,28                      | 62,29 | 68,77 | 71,44 | 75,50 | 79,19 | 82,58 |
| 30                              | 47,62      | 62,86                      | 63,87 | 70,35 | 73,02 | 77,08 | 80,77 | 84,16 |
| 31                              | 49,21      | 64,45                      | 65,46 | 71,94 | 74,61 | 78,67 | 82,36 | 85,75 |
| 32                              | 50,80      | 66,04                      | 67,05 | 73,53 | 76,20 | 80,26 | 83,95 | 87,34 |



| Length code<br>No. <sup>a</sup>                                                            | G<br>±0,13 | Length (G + B ref.) ± 0,25 |        |        |        |        |        |        |
|--------------------------------------------------------------------------------------------|------------|----------------------------|--------|--------|--------|--------|--------|--------|
|                                                                                            |            | 9                          | 10     | 12     | 14     | 16     | 18     | 20     |
| 34                                                                                         | 53,98      | 69,22                      | 70,23  | 76,71  | 79,38  | 83,44  | 87,13  | 90,52  |
| 36                                                                                         | 57,15      | 72,39                      | 73,40  | 79,88  | 82,55  | 86,61  | 90,30  | 93,69  |
| 38                                                                                         | 60,32      | 75,56                      | 76,57  | 83,05  | 85,72  | 89,78  | 93,47  | 96,86  |
| 40                                                                                         | 63,50      | 78,74                      | 79,75  | 86,23  | 88,90  | 92,96  | 96,65  | 100,04 |
| 42                                                                                         | 66,68      | 81,92                      | 82,93  | 89,41  | 92,08  | 96,14  | 99,83  | 103,22 |
| 44                                                                                         | 69,85      | 85,09                      | 86,10  | 92,58  | 95,25  | 99,31  | 103,00 | 106,39 |
| 46                                                                                         | 73,02      | 88,26                      | 89,27  | 95,75  | 98,42  | 102,48 | 106,17 | 109,56 |
| 48                                                                                         | 76,20      | 91,44                      | 92,45  | 98,93  | 101,60 | 105,66 | 109,35 | 112,75 |
| 50                                                                                         | 79,38      | 94,62                      | 95,63  | 102,11 | 104,78 | 108,84 | 112,53 | 115,92 |
| 52                                                                                         | 82,55      | 97,79                      | 98,80  | 105,28 | 107,95 | 112,01 | 115,70 | 119,09 |
| 54                                                                                         | 85,72      | 100,96                     | 101,97 | 108,45 | 111,12 | 115,18 | 118,87 | 122,26 |
| 56                                                                                         | 88,90      | 104,14                     | 105,15 | 111,63 | 114,30 | 118,36 | 122,05 | 125,44 |
| 58                                                                                         | 92,08      | 107,32                     | 108,33 | 114,81 | 117,48 | 121,54 | 125,23 | 128,62 |
| 60                                                                                         | 95,25      | 110,49                     | 111,50 | 117,98 | 120,65 | 124,71 | 128,40 | 131,79 |
| 62                                                                                         | 98,43      | 113,67                     | 114,69 | 121,16 | 123,83 | 127,89 | 131,58 | 134,97 |
| 64                                                                                         | 101,60     | 116,84                     | 117,86 | 124,33 | 127,00 | 131,06 | 134,75 | 138,62 |
| 66                                                                                         | 104,78     | 120,02                     | 121,04 | 127,51 | 130,18 | 134,24 | 137,93 | 141,32 |
| 68                                                                                         | 107,95     | 123,19                     | 124,21 | 130,68 | 133,35 | 137,41 | 141,10 | 144,49 |
| 70                                                                                         | 111,13     | 126,37                     | 127,39 | 133,86 | 136,53 | 140,59 | 144,28 | 147,67 |
| 72                                                                                         | 114,30     | 129,54                     | 130,56 | 137,03 | 139,70 | 143,76 | 147,45 | 150,84 |
| 74                                                                                         | 117,48     | 132,72                     | 133,74 | 140,21 | 142,88 | 146,94 | 150,63 | 154,02 |
| 76                                                                                         | 120,65     | 135,89                     | 136,91 | 143,38 | 146,05 | 150,11 | 153,80 | 157,19 |
| 78                                                                                         | 123,83     | 139,07                     | 140,09 | 146,56 | 149,23 | 153,29 | 156,98 | 160,37 |
| 80                                                                                         | 127,00     | 142,24                     | 143,26 | 149,73 | 152,40 | 156,46 | 160,15 | 163,54 |
| <sup>a</sup> Intermediate grip lengths may be purchased in 1,59 mm increment if necessary. |            |                            |        |        |        |        |        |        |

Table 5 — Oversize dimensions 1<sup>st</sup> Oversize

Dimensions in millimetres

| Thread <sup>a</sup><br>UNJF-3A<br>modified<br>(inch)                          | Oversize shank <sup>b</sup> |              |           |                   |                  | Mass Ref. (g)   |          |                       |             |          |                       |
|-------------------------------------------------------------------------------|-----------------------------|--------------|-----------|-------------------|------------------|-----------------|----------|-----------------------|-------------|----------|-----------------------|
|                                                                               | Dia.<br>code<br>No.         | Nom.<br>dia. | B<br>Ref. | D                 |                  | Head and thread |          |                       | Smooth part |          |                       |
|                                                                               |                             |              |           | T, B or P<br>code | Other<br>code    | Steel           | Titanium | Ni-<br>based<br>alloy | Steel       | Titanium | Ni-<br>based<br>alloy |
| 0.1900-32                                                                     | 3X                          | 5,16         | 7,62      | 5,146<br>5,134    | 5,146<br>5,121   | 1,60            | 0,87     | 1,72                  | 0,24        | 0,14     | 0,26                  |
| 0.2160-28                                                                     | 3AX                         | 5,95         | 8,00      | 5,939<br>5,927    | 5,939<br>5,914   | –               | 1,25     | 2,32                  | –           | 0,18     | 0,33                  |
| 0.2500-28                                                                     | 4X                          | 6,75         | 8,38      | 6,734<br>6,722    | 6,734<br>6,708   | 2,98            | 1,53     | 3,21                  | 0,41        | 0,23     | 0,45                  |
| 0.3125-24                                                                     | 5X                          | 8,33         | 9,91      | 8,321<br>8,309    | 8,321<br>8,296   | 5,10            | 2,81     | 5,49                  | 0,64        | 0,37     | 0,69                  |
| 0.3750-24                                                                     | 6X                          | 9,92         | 10,92     | 9,909<br>9,897    | 9,909<br>9,883   | 8,77            | 4,80     | 9,44                  | 0,92        | 0,53     | 0,99                  |
| 0.4375-20                                                                     | 7X                          | 11,51        | 12,57     | 11,496<br>11,484  | 11,496<br>11,471 | 13,63           | 7,19     | 14,67                 | 1,24        | 0,71     | 1,34                  |
| 0.5000-20                                                                     | 8X                          | 13,10        | 13,58     | 13,084<br>13,072  | 13,084<br>13,058 | 19,81           | 10,33    | 21,31                 | 1,61        | 0,93     | 1,73                  |
| 0.5625-18                                                                     | 9X                          | 14,68        | 15,49     | 14,658<br>14,646  | 14,658<br>14,633 | 29,22           | 16,12    | 31,44                 | 2,02        | 1,17     | 2,18                  |
| 0.6250-18                                                                     | 10X                         | 16,27        | 16,51     | 16,246<br>16,234  | 16,246<br>16,220 | 40,31           | 21,82    | 43,37                 | 2,50        | 1,45     | 2,70                  |
| 0.7500-16                                                                     | 12X                         | 19,45        | 22,98     | 19,421<br>19,409  | 19,421<br>19,396 | 97,08           | 55,30    | 104,45                | 3,59        | 2,05     | 3,87                  |
| 0.8750-14                                                                     | 14X                         | 22,62        | 25,65     | 22,596<br>22,584  | 22,596<br>22,571 | 152,00          | 86,58    | 163,54                | 4,88        | 2,78     | 5,24                  |
| 1.0000-12                                                                     | 16X                         | 25,80        | 29,72     | 25,771<br>25,759  | 25,771<br>25,746 | 225,00          | 128,16   | 242,09                | 6,36        | 3,63     | 6,85                  |
| 1.1250-12                                                                     | 18X                         | 28,97        | 33,40     | 28,945<br>28,933  | 28,945<br>28,920 | 326,00          | 149,94   | 278,21                | 8,02        | 4,56     | 8,47                  |
| 1.2500-12                                                                     | 20X                         | 32,15        | 36,79     | 32,121<br>32,109  | 32,121<br>32,097 | 379,14          | 214,51   | 398,02                | 9,95        | 5,63     | 10,45                 |
| <sup>a</sup> Thread according to AS8879 except diameter <i>TD</i> .           |                             |              |           |                   |                  |                 |          |                       |             |          |                       |
| <sup>b</sup> For diameter code 3X, shank diameter is incremented by 0.333 mm. |                             |              |           |                   |                  |                 |          |                       |             |          |                       |

Table 6 — Oversize dimensions 2<sup>nd</sup> Oversize

Dimensions in millimetres

| Thread <sup>a</sup><br>UNJF-3A<br>modified<br>(inch)                | Oversize shank      |              |                |           |                   |                  | Mass Ref. (g)   |                        |                       |                |                        |                       |
|---------------------------------------------------------------------|---------------------|--------------|----------------|-----------|-------------------|------------------|-----------------|------------------------|-----------------------|----------------|------------------------|-----------------------|
|                                                                     | Dia.<br>code<br>No. | Nom.<br>dia. | A              | B<br>Ref. | D                 |                  | Head and thread |                        |                       | Smooth part    |                        |                       |
|                                                                     |                     |              |                |           | T, B or P<br>code | Other code       | Steel<br>alloy  | Tita-<br>nium<br>alloy | Ni-<br>based<br>alloy | Steel<br>alloy | Tita-<br>nium<br>alloy | Ni-<br>based<br>alloy |
| 0.2500-28                                                           | 4Y                  | 7,14         | 11,68<br>11,05 | 8,38      | 7,130<br>7,118    | 7,130<br>7,104   | 2,98            | 1,53                   | 3,21                  | 0,44           | 0,25                   | 0,47                  |
| 0.3125-24                                                           | 5Y                  | 8,73         | 13,21<br>12,45 | 9,91      | 8,717<br>8,705    | 8,717<br>8,692   | 5,10            | 2,81                   | 5,49                  | 0,67           | 0,39                   | 0,73                  |
| 0.3750-24                                                           | 6Y                  | 10,32        | 15,75<br>14,86 | 10,92     | 10,305<br>10,293  | 10,305<br>10,279 | 8,77            | 4,80                   | 9,44                  | 0,95           | 0,55                   | 1,03                  |
| 0.4375-20                                                           | 7Y                  | 11,91        | 17,65<br>16,76 | 12,57     | 11,892<br>11,880  | 11,892<br>11,867 | 13,63           | 7,19                   | 14,67                 | 1,29           | 0,74                   | 1,38                  |
| 0.5000-20                                                           | 8Y                  | 13,49        | 20,07<br>19,18 | 13,58     | 13,480<br>13,468  | 13,480<br>13,454 | 19,81           | 10,33                  | 21,31                 | 1,66           | 0,96                   | 1,79                  |
| 0.5625-18                                                           | 9Y                  | 15,08        | 22,86<br>21,97 | 15,49     | 15,055<br>15,043  | 15,055<br>15,029 | 29,22           | 16,12                  | 31,44                 | 2,08           | 1,20                   | 2,24                  |
| 0.6250-18                                                           | 10Y                 | 16,67        | 24,64<br>23,75 | 16,51     | 16,642<br>16,630  | 16,642<br>16,617 | 40,31           | 21,82                  | 43,37                 | 2,56           | 1,48                   | 2,76                  |
| 0.7500-16                                                           | 12Y                 | 19,84        | 30,10<br>29,08 | 22,98     | 19,817<br>19,805  | 19,817<br>19,792 | 97,08           | 55,30                  | 104,45                | 3,67           | 2,09                   | 3,95                  |
| 0.8750-14                                                           | 14Y                 | 23,02        | 34,29<br>33,27 | 25,65     | 22,992<br>22,980  | 22,992<br>22,967 | 152,00          | 86,58                  | 163,54                | 4,96           | 2,83                   | 5,33                  |
| 1.0000-12                                                           | 16Y                 | 26,19        | 38,86<br>37,85 | 29,72     | 26,167<br>26,155  | 26,167<br>26,142 | 225,00          | 128,16                 | 242,09                | 6,46           | 3,68                   | 6,95                  |
| 1.1250-12                                                           | 18Y                 | 29,37        | 43,56<br>42,55 | 33,40     | 29,342<br>29,330  | 29,342<br>29,317 | 326,00          | 149,94                 | 278,21                | 8,13           | 4,63                   | 8,58                  |
| 1.2500-12                                                           | 20Y                 | 32,54        | 49,77<br>48,77 | 36,79     | 32,517<br>32,505  | 32,517<br>32,493 | 379,14          | 214,51                 | 398,02                | 10,08          | 5,70                   | 10,58                 |
| <sup>a</sup> Thread according to AS8879 except diameter <i>TD</i> . |                     |              |                |           |                   |                  |                 |                        |                       |                |                        |                       |

Table 7 — Oversize dimensions 3<sup>rd</sup> Oversize

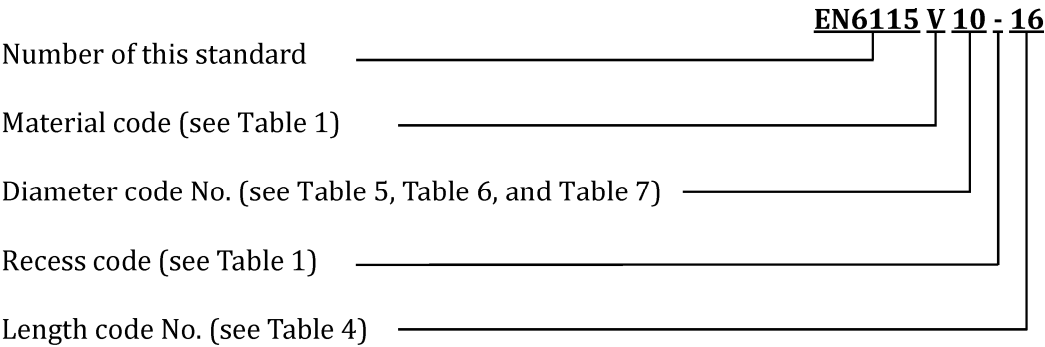
Dimensions in millimetres

| Thread <sup>a</sup><br>UNJF-3A<br>modified<br>(inch)                | Oversize shank      |              |                |           |                   |                  | Mass Ref. (g)   |                   |                   |                |                   |                   |
|---------------------------------------------------------------------|---------------------|--------------|----------------|-----------|-------------------|------------------|-----------------|-------------------|-------------------|----------------|-------------------|-------------------|
|                                                                     | Dia.<br>code<br>No. | Nom.<br>dia. | A              | B<br>Ref. | D                 |                  | Head and thread |                   |                   | Smooth part    |                   |                   |
|                                                                     |                     |              |                |           | T, B or P<br>code | Other<br>code    | Steel<br>alloy  | Titanium<br>alloy | Ni-based<br>alloy | Steel<br>alloy | Titanium<br>alloy | Ni-based<br>alloy |
| 0.6250-18                                                           | 10Z                 | 17,46        | 25,27<br>24,38 | 16,76     | 17,437<br>17,424  | 17,437<br>17,412 | 40,31           | 21,82             | 43,37             | 2,68           | 1,55              | 2,89              |
| 0.7500-16                                                           | 12Z                 | 20,64        | 30,63<br>29,61 | 23,23     | 20,612<br>20,599  | 20,612<br>20,587 | 97,08           | 55,30             | 104,45            | 3,81           | 2,18              | 4,11              |
| 0.8750-14                                                           | 14Z                 | 23,81        | 34,84<br>33,82 | 25,90     | 23,787<br>23,775  | 23,787<br>23,762 | 152,00          | 86,58             | 163,54            | 5,13           | 2,93              | 5,52              |
| 1.0000-12                                                           | 16Z                 | 26,99        | 39,42<br>38,41 | 29,96     | 26,962<br>26,950  | 26,962<br>26,937 | 225,00          | 128,16            | 242,09            | 6,65           | 3,79              | 7,16              |
| <sup>a</sup> Thread according to AS8879 except diameter <i>TD</i> . |                     |              |                |           |                   |                  |                 |                   |                   |                |                   |                   |

5 Designation

This type of standard shall be designated according to the philosophy of the following example:

EXAMPLE



If the diameter code ends with an letter (e. g. 3A, 3X, 3AX, 4X or 4Y) do not use the “-” recess code (e. g. EN6115K3A5, EN6115-3A5, EN6115B3E5, EN6115B3AXE5).

6 Marking

Parts shall be marked according to EN 2424, style B. the marking shall be recessed with maximum depth of 0,25 mm.

7 Technical specification

The technical specification shall be according to EN 6116.

## **8 Quality assurance**

The manufacturer's operations shall be an approved production organisation for aerospace products and shall demonstrate that it has implemented and is able to maintain a quality management system (e. g. according to EN 9100 or an equivalent aerospace accepted and established quality management system).

The qualification procedure for aerospace standard products (e. g. according to EN 9133 or an equivalent aerospace accepted and established qualification procedure) shall be used and documented according to the specified tests if not otherwise agreed between customer and supplier.

## Annex A (informative)

### Record of revisions

**Table A.1 — Record of revisions**

| Edition    | Clause modified                                                                | Description of modification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1<br>06/02 |                                                                                | New standard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3<br>07/02 | § 2<br>Tables 3, 4 and 5                                                       | Normative references updated.<br>Introduction of diameter code Nos 20 and 22.<br>Data for P max.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3<br>08/03 | § 3.1<br>Table 2                                                               | Limitation of application: Large diameters (-12 to -18) titanium fasteners.<br>Addition of mechanical characteristics for -12 Inconel fasteners, update of tensile strength for -4 Inconel fastener.                                                                                                                                                                                                                                                                                                         |
| 4<br>07/04 | Table 2<br>Table 4<br>Table 5                                                  | Mechanical requirements added for -20.<br>Limitation of grip length<br>Oversize code Z added.                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5<br>12/07 | 3.2<br>Table 1<br>Table 2<br>Figure 1<br><br>Table 2<br>Table 3<br><br>Table 5 | New references added.<br>Code B added. References updated.<br>Inconel mechanical properties updated for dia. code No. 9.<br>Drawing with oversize's details updated.<br>Note 4 added for code B.<br>Note 1 modified.<br>Mechanical properties of dash 3A for Inconel added<br>Min. tensile strength of dash 8 updated.<br>Headings and masses updated.<br>Q modified.<br>Mass added for oversize's dimensions.<br>Dimensions of Ø A modified for dia. code No. 7Y: 17,65/17,27 mm changed to 17,65/16,76 mm. |
| 6<br>11/09 | 2<br>Table 1 and table 3                                                       | Normative references updated. Reference AMS QQ-P-35 replaced by AMS2700.<br>Addition of P code                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7<br>08/10 | Fig. 1<br>Table 1, Fig.1                                                       | 5 Lobe high torque recess version added<br>Table 1 and fig. 1 updated, footnote 5 added and footnote 4B updated                                                                                                                                                                                                                                                                                                                                                                                              |
|            | Table 1<br><br>Table 3                                                         | Footnote updated.<br>5 Lobe high torque recess with recess code E added for code B pins<br>Dimension P max. moved to column Hexagonal recess<br>Dimensions for 5 Lobe high torque recess added<br>Dimension T min. and P max. updated                                                                                                                                                                                                                                                                        |

| <b>Edition</b> | <b>Clause modified</b> | <b>Description of modification</b>                                                                                                                                                                                                                                               |
|----------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | Table 5                | Column D amended with "T, B or P code"                                                                                                                                                                                                                                           |
|                | Designation            | EN6115B3AX5 modified to EN6115B3AXE5<br>Recess code added without influence of existing designation.<br>Footnote a) added.                                                                                                                                                       |
| 8<br>06/14     | Table 1                | Finish code K has been replaced by finish code F according to the ACF requirements.                                                                                                                                                                                              |
|                | Table 2                | Diameter ranges corrected in line with Table 2.<br>Addition of Minimum double shear, Minimum tensile strength and Maximum Fatigue load for Inconel material dash number 16.                                                                                                      |
| 9<br>03/18     | 3<br>Table 1           | Limitation of application removes as it is a design requirement<br>For Codes K, B and L Finish amended to read EN4473 Type II<br>Footnote 2) added.                                                                                                                              |
|                | 3.5<br>Table 3         | Footnote for thread rolling of steel parts removed<br>Dimension Z amended to read Z max<br>ØY dimension for diameter 2 dimensions amended in accordance with manufacturers drawings.<br>Dimension J for diameters 6, 8 and 18 amended in accordance with manufacturers drawings. |
| 10             |                        |                                                                                                                                                                                                                                                                                  |

**prEN 6115:2019 (E)**

## **Bibliography**

EN 9100, *Quality Management Systems — Requirements for Aviation, Space and Defence Organizations*

EN 9133, *Aerospace series — Quality Management Systems — Qualification Procedure for Aerospace Standard Products*